Beacon Outfitters | COO decision memo

To: Merchandising VP | From: Anshuman, Inka, Juan Diego
November 3, 2026 | ISOM 839 academic case

Approve a targeted replenishment plan

Approve the four-week plan: 512 cases (4,028 units) on 27 store visits. Expected contribution profit is $213,503.10, $177,903.35 above no deliveries and $6,123.90 above the feasible forecast-driven rule.

Secure 100 additional J200 down parkas for week 1 if the incremental procurement premium is below the $10,169.20 gross gain. Use the store-specific calendar rather than a blanket every-other-week rule, which costs $3,124.00 in profit.

With weeks 3–4 demand 20% lower, replan for $196,074.61; keeping the base shipments earns $192,979.83. With demand 20% higher, replan for $228,223.15; the fixed base plan earns $221,594.99.

The plan sells 4,598 units, fills 96.13% of demand and leaves 185 units of unmet demand. The detailed case-by-case shipment file accompanies the notebook; the calendar below gives each store's delivery weeks.

Store	Delivery weeks	Units
S01	1, 2, 4	562
S02	1, 2, 4	406
S03	1, 2, 4	370
S04	1, 2, 4	330
S05	1, 3	376
S06	1, 3	334
S07	1, 4	280
S08	1, 4	248
S09	1, 3	326
S10	1, 3	292
S11	1, 3	282
S12	2	222

The money and the scarce stock

Policy	Profit	Visits	Fill rate
base	$213,503.10	27	96.13%
do_nothing	$35,599.75	0	15.76%
naive	$207,379.20	48	95.25%
alternate_week	$210,379.10	24	94.36%

Contribution = $221,013.00 sales margin minus $1,399.90 shelf holding and $6,110.00 truck costs. Unit cost is counted once in margin, including sales from opening stock as specified in the case.

SKU (+100 units at start of week 1)	Profit gain
J200	$10,169.20
H600	$1,019.80
G700	$953.90
P500	$13.60
F300	$0.00
J100	$0.00
B400	$0.00
S800	$0.00

A binding supply limit alone does not identify the best purchase. J200 has the largest finite profit opportunity. The values above include pack sizes, shelf limits and changed delivery decisions; they are not per-unit shadow prices.

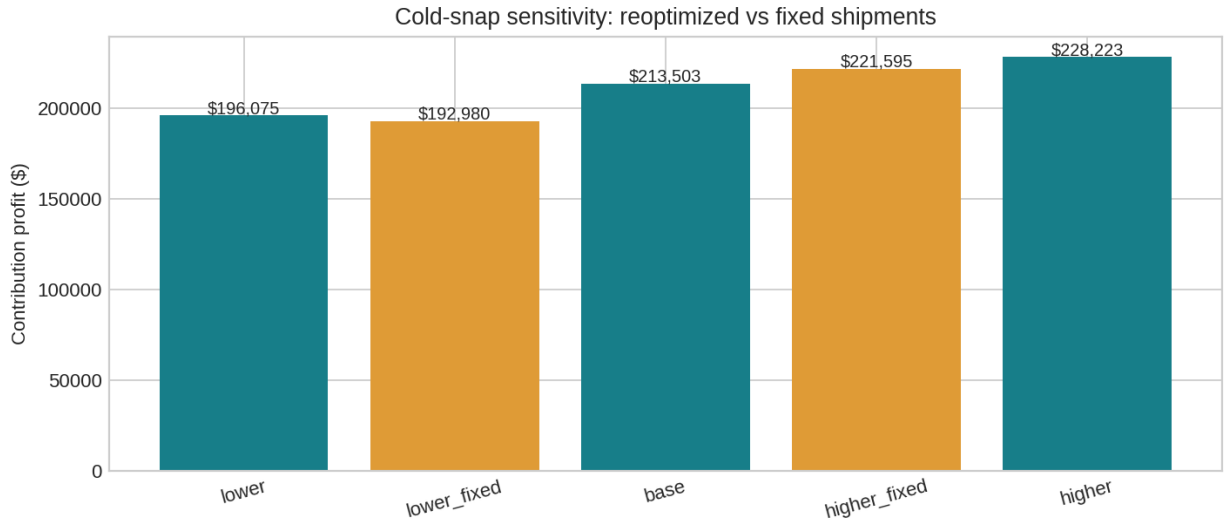
Naive comparison rule

Each week, process stores by ID and SKUs by descending unit margin. Request enough cases for that week and cap allocation by available DC stock, shelf room and truck capacity. Sales and carryover adjust with fixed shipments. This feasible benchmark is intentionally simple; it is not claimed to reproduce an actual current company policy.

The forecast changes the best plan

With weeks 3–4 demand 20% lower, replan for $196,074.61; keeping the base shipments earns $192,979.83. With demand 20% higher, replan for $228,223.15; the fixed base plan earns $221,594.99.

Replanning instead of holding the base deliveries improves lower-demand profit by $3,094.78 and higher-demand profit by $6,628.16. Higher demand increases the best attainable profit here; it does not automatically destroy profit.



Truck recommendation

Burlington VT (S08, smallest shelf) costs $3.06 per delivered unit; Boston Back Bay (S01, largest shelf) costs $0.96. Truck capacity is 420 units in both; cost efficiency depends on actual volume and visits. Keep store-specific consolidation. A forced odd/even-week-only policy saves visits but sacrifices $3,124.00 net profit.

Technical evidence and limits

Gurobi and CP-SAT both prove the same base optimum; all reported mandatory reoptimizations are proven optimal. Independent checks cover flow, demand, shared shelf after arrival, truck activation, cumulative DC, integer packs and recomputed profit. Currency uses cents; CP-SAT uses fifth-units so fractional scenario demand is exact. Opening stock follows the published margin convention. No routing, terminal salvage, safety stock or demand learning is assumed.

Team action: review and defend every constraint, verify private Canvas rules, rehearse the five-minute briefing, and publish the reviewed notebook only after authorization. Local notebook execution does not establish a hosted Google Colab test. The Word brief was absent; public course case/syllabus source was read from GitHub because the website was blocked.